

Q1. Ionization energy (IE) of an one-electron atom is the minimum energy required to remove the electron from the lowest energy state of the atom. Using Bohr's theory, find the radius (in atomic unit a_0) of the 1st Bohr orbit, and ionization energy (in atomic unit E_h) for the following atoms/ions:

Properties	H	He ⁺	Li ²⁺	Be ³⁺	B ⁴⁺
r [a_0]					
IE [E_h]					

Interpret your results.

PROPERTIES	H	He ⁺	Li ²⁺	Be ³⁺	B ⁴⁺
r [a_0]					
IE [E_h]					

$$r = \frac{n^2}{Z} a_0$$

atomic units
of length,
read as bohr

$$r_H = 1 a_0$$

$$r_{He^+} = \frac{1}{2} a_0$$

$$r_{Li^{2+}} = \frac{1}{3} a_0$$

$$r_{Be^{3+}} = \frac{1}{4} a_0$$

$$r_{B^{4+}} = \frac{1}{5} a_0$$

A1

For
 $n=1$

$$IE = \Delta E = E_{UB} - E_1$$

E_{UB} : Energy of e^- at unbound state

$$E = - \frac{Z^2}{2n^2} E_h$$

atomic units of energy, hartree

E_1 : Energy of the lowest state

$n=1$

$$E_1^H = - \frac{1}{2} E_h$$

$$E_1^{He^+} = - \frac{2^2}{2} E_h$$

$$E_1^{Li^{2+}} = - \frac{3^2}{2} E_h$$

$$E_1^{Be^{3+}} = - \frac{4^2}{2} E_h$$

$$E_1^{B^{4+}} = - \frac{5^2}{2} E_h$$

E_{UB} Unbound or ionized when e^- & nucleus are completely separated i.e., $E_{UB} = 0$

PROPERTIES	H	He ⁺	Li ²⁺	Be ³⁺	B ⁴⁺
$r [a_0]$	1	1/2	1/3	1/4	1/5
$IE [E_h]$	1/2	4/2	9/2	16/2	25/2

As Z , atomic no. ↑

Bohr radius ↓

As Z , atomic no. ↑

Ionization energy ↑

Q2. Which of the following operator-function combinations would yield eigenvalue equation? Find the eigenvalue of the eigenfunction.

(i) $\hat{O} = \frac{d^2}{dx^2}$ and $f(x) = \cos\left(\frac{x}{4}\right)$

(ii) $\hat{O} = \frac{d}{dx}$ and $f(x) = e^{-3.4x^2}$

(iii) $\hat{O} = \frac{d}{dx}$ and $f(x) = \cos(ax)$

(iv) $\hat{O} = \frac{d}{dt}$ and $f(t) = e^{i\omega t}$

(v) $\hat{O} = \frac{\partial}{\partial y}$ and $f(x, y) = xy^3$

(vi) $\hat{O} = \frac{d^2}{dx^2} + 2\frac{d}{dx} + 3$ and $f(x) = e^{ax}$

A2

$\hat{O}\Psi(x) = O\Psi(x)$

operator ← function → Constant value

(i) $\frac{d^2}{dx^2} \left(\cos \frac{x}{4} \right)$

$\Rightarrow \frac{d}{dx} \left[\frac{d}{dx} \left(\cos \frac{x}{4} \right) \right]$

$\Rightarrow \frac{d}{dx} \left[-\sin \frac{x}{4} \right] \times \frac{1}{4}$

$\Rightarrow -\left(\cos \frac{x}{4} \right) \times \frac{1}{4} \times \frac{1}{4}$

$\Rightarrow -\frac{1}{16} \left(\cos \frac{x}{4} \right)$ $\hat{O}\Psi = O\Psi$ ✓

Eigenvalue

(ii) $\frac{d}{dx} (e^{-3.4x^2})$ $-3.4x^2$

$\Rightarrow -3.4 \times 2 \times x \times e$

$\hat{O}\Psi \neq O\Psi$

(iii) $\frac{d}{dx} (\cos ax)$

$\Rightarrow -a \sin ax$

$\hat{O}\Psi \neq O\Psi$

(iv) $\frac{d}{dt} \left[e^{i\omega t} \right]$

$\Rightarrow i\omega x e^{i\omega t}$

Eigenvalue

$\hat{O}\Psi = O\Psi$

(v) $\frac{\partial}{\partial y} (f(x, y))$

$\Rightarrow \frac{\partial}{\partial y} (xy^3) = 3xy^2$

$\hat{O}\Psi \neq O\Psi$

(vi) $\left[\frac{d^2}{dx^2} + 2\frac{d}{dx} + 3 \right] (e^{ax})$

$\Rightarrow a^2 e^{ax} + 2a e^{ax} + 3e^{ax}$

$\Rightarrow (a^2 + 2a + 3) e^{ax}$

Eigenvalue

$\hat{O}\Psi = O\Psi$

A3

Q3. Show that $\Psi(x) = e^{ikx} + e^{-ikx}$ is an eigenfunction of the Hamiltonian operator $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ for a free-particle in one-dimension. What is the eigenvalue? What is the observable here? Is $\Psi(x)$ also an eigenfunction of the linear momentum operator $\hat{p}_x$?

$$\Psi(x) = e^{ikx} + e^{-ikx}$$

$$\hat{H}\Psi(x) = \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} (e^{ikx} + e^{-ikx}) \right]$$

$$= -\frac{\hbar^2}{2m} \left[ik \times ik \times e^{ikx} + (-ik) \times (-ik) \times e^{-ikx} \right]$$

$$= -\frac{\hbar^2}{2m} \left[(ik)^2 \left\{ e^{ikx} + e^{-ikx} \right\} \right]$$

$$= -\frac{\hbar^2}{2m} \times i^2 k^2 \Psi(x)$$

$$= -\frac{\hbar^2}{2m} (-1) k^2 \Psi(x)$$

$$= \frac{\hbar^2 k^2}{2m} \Psi(x)$$

Eigenvalue, observable is total energy

$$\hat{p}_x = -i\hbar \frac{d}{dx}$$

$$\hat{p}_x \Psi(x) = -i\hbar \frac{d}{dx} [e^{ikx} + e^{-ikx}]$$

$$= -i\hbar \left[(ik) \times e^{ikx} + (-ik) \times e^{-ikx} \right]$$

$$= -i\hbar \left[(ik) \left\{ e^{ikx} - e^{-ikx} \right\} \right]$$

$$= +\hbar k \left[\underbrace{e^{ikx} - e^{-ikx}}_{\neq \Psi(x)} \right]$$

So, $\Psi(x)$ here, not an eigenfunction of $\hat{p}_x$

Q1 Show that $f(x, y, z) = (\sin ax)(\sin by)(\sin cz)$ is an eigenfunction of the following operator (known as Laplacian operator):

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

A4 $f(x, y, z) = (\sin ax)(\sin by)(\sin cz)$

$$\hat{\nabla}^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$$

$$\hat{\nabla}^2 f(x) = \underbrace{\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)}_{\hat{\nabla}^2} \underbrace{(\sin ax)(\sin by)(\sin cz)}_{f(x)}$$

$$= \left[(\sin by)(\sin cz) \frac{d^2}{dx^2} (\sin ax) + (\sin ax)(\sin cz) \frac{d^2}{dy^2} (\sin by) + (\sin ax) \sin(by) \frac{d^2}{dz^2} (\sin cz) \right]$$

$$= \left[(\sin by)(\sin cz) (-a^2) (\sin ax) + (\sin ax)(\sin cz) (-b^2) (\sin by) + (\sin ax)(\sin by) (-c^2) \sin(cz) \right]$$

$$= \underbrace{-(a^2 + b^2 + c^2)}_{\text{Eigenvalue}} \underbrace{(\sin ax)(\sin by)(\sin cz)}_{f(x)}$$

Q5. Indicate which of the following functions are "acceptable" wavefunctions. Give reasons.

(i) $f(x) = x^2 + 1$, where x can have any value (ii) $f(x) = \pm\sqrt{x}$, where $x \geq 0$

(iii) $f(x) = \frac{1}{\sqrt{2}} \sin\left(\frac{x}{2}\right)$, where $\left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right)$

(iv) $f(x) = e^{-x^2}$, where $0 \leq x < \infty$

A5

① $f(x) = x^2 + 1$, where x can have any value
As x approaches positive or negative ∞ , ψ also approaches infinity. Hence, NOT ACCEPTABLE

② $f(x) = \pm\sqrt{x}$ $x \geq 0$

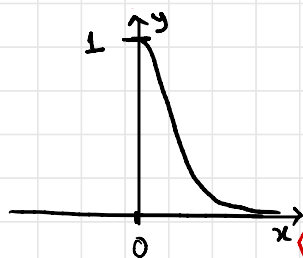
$f(x)$ has two values for a particular x
Not single-valued, not finite, hence NOT ACCEPTABLE

④ $f(x) = e^{-x^2}$ $0 \leq x < \infty$

Finite, Single-valued, Continuous

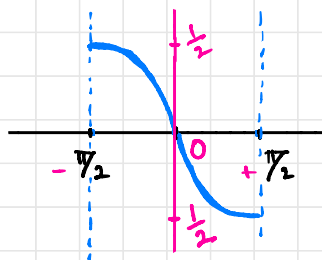
✓ Continuous slope (i.e., the function is differentiable) for given x values

ACCEPTABLE



$e^{-x^2} = \frac{1}{e^{x^2}}$
at $x \rightarrow \infty$
 $f(x) \rightarrow 0$
at $x = 0$
 $f(x) = 0$

③ $f(x) = \frac{1}{\sqrt{2}} \sin\left(\frac{x}{2}\right)$ $\left\{-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right\}$



ACCEPTABLE
* List all four conditions